



LOGICAL REASONING

Directions [Q. Nos. 1-4] A circular field with inner radius of 10 m and outer radius of 20 m is divided into 5 successive stage for ploughing. The ploughing at each stage, with starting points P1, P2, P3, P4 and P5 was allotted to one of the five farmers F1, F2, F3, F4 and F5 not necessarily in that order.

- F5 was allotted the stage starting a point P4.
 - The stage from P5 to P3 was not the first stage.
 - F4 was allotted the work of the fourth stage.
 - Finishing point of stage 3 was P1 and the work was not allotted to F1.
 - F3 was allotted the work of stage ending at P5.
1. Which of the following is the finish point for farmer F2?
(a) P1 (b) P2
(c) P3 (d) P4
 2. Which stage was ploughed by F5
(a) 2 (b) 3
(c) 4 (d) 5
 3. What are the starting and ending points of the field ploughed by F4?
(a) P1 and P2 (b) P1 and P4
(c) P4 and P2 (d) P2 and P4
 4. What is the starting point for stage 3?
(a) P2 (b) P3
(c) P4 (d) Cannot be determined
 5. If Tuesday falls on the fourth of a month, then which day will fall three days after 24th of the same month?
(a) Monday (b) Tuesday
(c) Thursday (d) Friday
 6. If the statements "All chickens are birds", "Some chickens are hens" and "Female birds lay eggs", are all facts, then which of the following must also be a fact?
I. All birds lay eggs
II. Some hens are birds
III. Some chickens are not hens
(a) I and II (b) II and III
(c) I and III (d) Neither I nor II nor III

Directions [Q. Nos. 7-10]

- There is a family of six members A, B, C, D, E and F.
 - There are two married couples in the family and the family members represent three generations.
 - Each member has a distinct choice of a colour amongst Green, Yellow, Black, Red, White and Pink.
 - No lady members likes either Green or White.
 - C, who likes Black colour, is the daughter-in-law of E.
 - B is the brother of F and son of D and likes Pink.
 - A is the grandmother of F and F does not like Red.
 - Wife of the husband having a choice for Green colour likes Yellow.
7. Which of the following is the colour preference of A?
(a) Red (b) Yellow
(c) Either Red or Yellow (d) Cannot be determined

8. Which of the following could be the colour combination of one of the couples?
(a) Yellow-Red (b) Green-Black
(c) Red-Yellow (d) Yellow-Green
9. Which of the following is one of the married couples?
(a) CD (b) AC
(c) AD (d) Cannot be determined
10. Which of the following is true about F?
(a) Brother of B (b) Sister of B
(c) Daughter of C (d) Cannot be determined
11. If the English word "EXAMINATION" is coded as 56149512965, then the word "GOVERNMENT" is coded as
(a) 7645954552 (b) 7654694562
(c) 7645955423 (d) 7654964526
12. Gopal starts from his house towards West. After walking a distance 30m, he turned towards right and walked 20m. He turned left and after moving a distance of 10m, turned to his left again and walked 40m. Then, he turned left and walked 5m. Finally, he turns to his left. In which direction is the walking now?
(a) North (b) South
(c) East (d) South East
13. Read the conclusion and then decide which of the given conclusions logically follows from the two given statements (i) and (ii) disregarding commonly known facts.
Statements:
(i) No women teacher can play.
(ii) Some women teachers are athletes.
Conclusions:
(i) Male athletes can play.
(ii) Some athletes can play.
(a) Only conclusion I follows
(b) Only conclusion II follows
(c) Either I or II follows
(d) Neither I nor II follows
14. Which of the following come next in the series 8, 6, 9, 23, 87,.....?
(a) 28 (b) 226 (c) 324 (d) 429
15. In an examination there are 100 questions divided into 3 parts A, B and C and each part should contain atleast one question. Each question in parts A, B and C carry 1, 2 and 3 marks respectively. Part A is for atleast 60% of the total marks and part B should contain 23 questions. How many questions must be set in part C?
(a) 1 (b) 2
(c) 3 (d) Cannot be determined
16. If $\div$ means addition, $-$ means division, $\times$ means subtraction and $+$ means multiplication, then the value of $\frac{(36 \times 4) - 8 \times 4}{4 + 8 \times 2 + 16 \div 1}$ is
(a) 0 (b) 8
(c) 12 (d) 16
17. Which letter is the word CYBERNETICS occupies the same position as it does in the English alphabet?
(a) C (b) E



(c)I (d)T

18. The remainder when 2^{31} is divided by 5 is

(a)1 (b)2
(c)3 (d)4

Directions[Q. Nos. 19-20] the letter of English alphabet from A to M were written, leaving space for one letter between every two letters and then the remaining letters were inserted beginning with N and ending the series with Z after M.

19. Which letter would be 3rd to the right of the 7th letter from the left?

(a) C (b) O
(c) R (d) S

20. Which letter would be exactly in the middle of eighteenth letter from the beginning and fifteenth from the end?

(a)G (b)H
(c)J (d)L

21. How many 3- digit numbers divisible by 5, can be formed using the digits 2, 3, 5, 6, 7 and 9, without repetition of digits?

(a)216 (b)20
(c)17 (d)18

22. Using only 2, 5, 10, 25 and 50 paise coins, what is the smallest number of coins required to pay exactly 78paise, 69 paise and Rs.1.01 to three different persons?

(a)19 (b)20
(c)17 (d)18

23. Which of the following two patterns will fit in the blanks of the series $ZA_5, Y_4B, XC_6, W_3D, \dots$?

(a) VE_7 and U_2E (b) V_2E and U_7F
(c) VE_7 and U_2F (d) VF_7 and U_2E

24. Which of the following numbers comes next in the two-digit decimal number sequence 61, 52, 63, 94,?

(a)65 (b)64
(c)56 (d)46

25. Three ladies X, Y and Z marry three man A, B and C, X is married to A, Y is not married to an engineer. Z is not married to a doctor, C is not a doctor and A is a lawyer. Then, which of the following statements is correct?

(a) X is married to a doctor
(b) Y is married to C, who is a doctor
(c) Z is married to B, who is an engineer
(d) None of the above

26. There are five books A, B, C, D and E placed on a table .If A is placed below E, C is placed D, B is placed below A and D is placed between A and E, then which of the following books can be on the top?

(a) D or E (b) C or E
(c)A or E (d)None of these

27. Among five children A, B, C, D and E, B is taller than E but shorter than D. A is shorter than C but taller than D. If all the children stand in a line according to their heights, then who would be the fourth if counted from the tallest one?

(a)D (b)C
(c)B (d)A

Directions[Q.Nos.28-31]

- In a family of six person A, B, C, D, E and F there are two married couples.
- D is grandfather of A and mother B.
- C is wife of B and mother F.
- F is the granddaughter of E.

28. What is C to A?

(a) Daughter (b)Grandmother
(c) Mother (d) Cannot be determined

29. How many male members are there in the family?

(a) Two (b) Three
(c) Four (d) Cannot be determined

30. Who among the following in one of the couples?

(a)CD (b)DE
(c)EB (d)Cannot be determined

31. Which of the following is true?

(a)A is brother of F (b)A is sister of F
(c)B has two daughters (d)None of these

Directions [Q. Nos. 32-35] A, B, C, D, E, F and G are seven girls having different amount of money from among Rs.10,20,40,60,80,120 and 200 with them . They had 3 chocolates, 2 toffees and 2 lollipops together, each other having one of these seven items.

B and F do not have chocolates and they have Rs.200 and Rs.80, respectively.

- C has Rs.60 with her and G has an amount which neither Rs.40 nor Rs.120.
- A has Rs10 and does not have a toffee.
- The girl having Rs.40 with her is the only one other than A to have the same type of item.
- E and girl having Rs.20 with her have the same find of item.

32. How much amount does G have with her?

(a) Rs.20 (b) Rs.10
(c) Rs.60 (d) None of these

33. Which of the following girl have chocolates with items?

(a) F, C, G (b) C, G, E
(c) C, G, D (d) G, D, E

34. Which of the following combination is definitely correct?

(a)C-chocolate Rs.60 (b)G- toffee-Rs.20
(c) C-chocolate Rs.40 (d)None of these

35. Which girls has Rs.40 with her?

(a)E (b)A
(c)D (d)None of these

36. P,Q,R,S,T,U and V are sitting in a row facing North. In order determine, who is sitting exactly in the middle of the row, which of the following information is needed?

- T and U are sitting at extreme ends of the row.
- S is third to the right of T



III. Q is four places to the left of R and P is two places of the left of V

- (a) I and II only are sufficient
(b) I and III only are sufficient
(c) I and either II or III are sufficient
(d) I, II and III

37. How many times do the hour and the minute hands of a clock overlap in 24 h?

- (a) 24 (b) 22
(c) 26 (d) 20

38. In a certain code, TOGETHER is coded as RQEGRJCT. In the same code, PAROLE will be written as

- (a) NCPQJG (b) NCQPJG
(c) RCPQJK (d) RCTQNG

39. A drawer contains 10 black and 10 brown socks which are all mixed up. What is the smallest number of socks to be taken from the drawer to decided without seeing them, to be sure that there is atleast one pair of socks of the same colour ?

- (a) 11 (b) 10
(c) 3 (d) Cannot be determined

40. Find the missing number in the series: 4, 7, 25, 10, , 20, 16, 19

- (a) 13 (b) 15
(c) 20 (d) 28

ANSWER

1.	A	2.	D	3.	B	4.	B	5.	C
6.	B	7.	B	8.	D	9.	A	10.	A
11.	A	12.	B	13.	D	14.	D	15.	A
16.	A	17.	C	18.	C	19.	C	20.	B
21.	B	22.	A	23.	C	24.	D	25.	D
26.	D	27.	C	28.	C	29.	D	30.	D
31.	D	32.	A	33.	B	34.	A	35.	C
36.	A	37.	B	38.	A	39.	C	40.	a

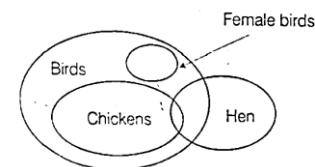
SOLUTION

Solution [Q. Nos. 1 to 4]

The given information can be represented as:

Stage	Farmer	Starting point of the stage	Ending point of the stage
First	F ₃	P ₂	P ₅
Second	F ₁	P ₅	P ₃
Third	F ₂	P ₃	P ₁
Fourth	F ₄	P ₁	P ₄
Fifth	F ₅	P ₄	P ₂

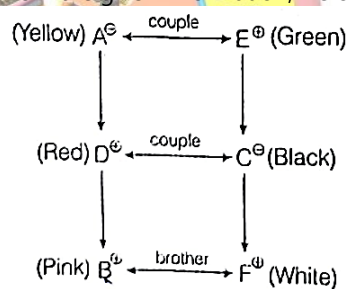
- (a) P is the finishing point of
- (d) 5th stage is ploughed by
- (b) P and P
- (b) P is the starting point of stage 3.
- (c) The 4th day is Tuesday, then 11th, 12th and 25th also Tuesdays.
Then, 3 days after 24th is 27th, which is Thursday.
- (b)



So, only conclusion II and III follow.

Solutions [Q. Nos. 7 to 10]

From the given information, we can draw a relation diagram,



- (b) A likes yellow colour.
- (d) Yellow-Hypen green is the colour combination of AE
- (a) CD is the married couple.
- (a) F is the brother of B.
- (a) As,

E	X	A	M	I	N	A	T	I	O	N
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
5	6	1	4	9	5	1	2	9	6	5

The pattern in the above word is as follows.

E = 5

X = 24 = 2 + 4 = 6

A = 1

M = 13 = 1 + 3 = 4





The number formed by reversing the digit of each number of the sequence is square of consecutive natural numbers starting from 4. See how,

$$61 \dots 16 = 4^2$$

$$52 \dots 25 = 5^2$$

$$63 \dots 36 = 6^2$$

$$94 \dots 49 = 7^2$$

Therefore, the required number = $8^2 = 64 = 46$.

25. (d) From the given information

Wife	Husband
X	A (Lawyer)
Y	B (Doctor)
Z	C (Engineer)

From the above table, no statement is correct.

26. (d) From the statements,
A is placed below E

E
A

C is placed below D

D
C

B is placed below A

A
B

D is placed between A and E, then

E
D
A
B
C

So, E will be on the top.

27. (c) Given that

B is taller than E but shorter than D.

$$D > B > E$$

A is shorter than C but taller than D.

$$D < A < C$$

Then the order is

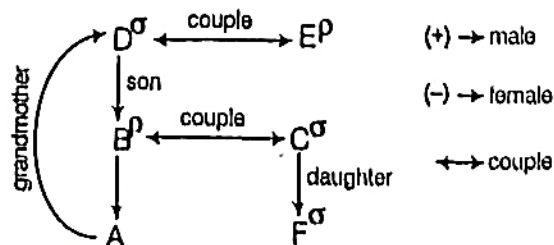
$$C > A > D > B > E$$

$$1 \quad 2 \quad 3 \quad 4 \quad 5$$

So, B is fourth.

Solutions [Q. Nos. 28 to 31]

From the given information in the question, we can draw a relation diagram,



28. (c) 'C' is the mother of A

29. (d) It cannot be determined because the gender of A is not defined.

30. (d) DF is the couple.

31. (d) No statement is true.

Directions [Q. Nos. 32-35]

The given information in the question can be summarised as:

Girl	Money	Item
A	10	Lollipops
B	200	Toffees
C	60	Chocolates
D	40	Lollipops
E	120	Chocolates
F	80	Toffees
G	20	Chocolates

32. (a) From the table, it is clear that G has Rs.20.

33. (b) From the table, it is clear that C, G and E has chocolates with them.

34. (a) Combination in option (a) is correct.

35. (c) From the table, it is clear that D has Rs.40 with her.

36. (a) From the statement I
1-T and U are sitting at extreme ends of the row.

Case I T U

Case II U I

From statement II

S is third to the right of T.

Then, from case I

T S U

So, S is sitting exactly in the middle.

So, only statement I and II are sufficient.

37. (b) The hands of a clock coincide 11 times in every 12 h (Since, between 11 and 1, they coincide only once i.e. at 12 O'clock).

AM	PM
12:00	12:00
1:05	1:05
2:11	2:11
3:16	3:16
4:22	4:22
5:27	5:27
6:33	6:33





7:38	7:38
8:44	8:44
9:49	9:49
10:55	10:55

The hands overlap about 65 min, not every 60 min. So, the hand coincide 22 times in a day.

38. (a) As,

As,
 $\begin{array}{cccccccc} T & O & G & E & T & H & E & R \\ -2 & +2 & -2 & +2 & -2 & +2 & -2 & +2 \\ \hline R & Q & E & G & R & J & C & T \end{array}$

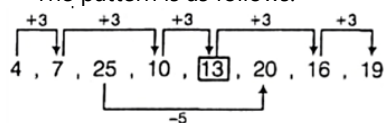
Similarly,

$\begin{array}{cccccc} P & A & R & O & L & E \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ N & C & P & Q & J & G \end{array}$

39. (c) Even if first 2 are different colours, 3rd one has to be black or brown. So, the smallest number of socks will be 3.

40. (a) 4, 7, 25, 10, 20, 16, 19

The pattern is as follows:



Two series alternate here, with every third number following a different patterns. In the main series, 3 is added to each number to arrive at the next. In the alternating series, 5 is subtracted from each number to arrive at the next.

